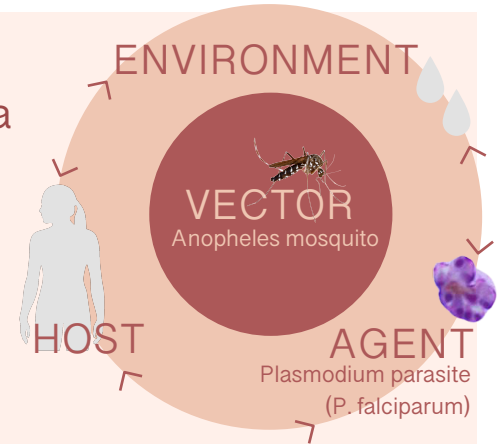


ECOLOGY OF MALARIA

Environmental drivers of malaria transmission in Africa

Malaria remains a major public-health challenge, causing nearly 600,000 deaths worldwide in 2023 (WHO). Transmission persists where mosquito and parasite biology align with local climate. Temperature controls parasite development inside the mosquito, while rainfall, humidity, and landscape shape where vectors can survive and breed. This interaction is summarised in the epidemiological triad on the right, showing the core components required for malaria transmission →

This map shows Africa's climatic suitability for malaria with data from July, overlaying temperature and rainfall reclassified into 0–3 suitability scores. July represents a peak transmission period due to the favourable climate. Risk concentrates in consistently warm tropical belts (it's winter in July in Southern Africa).



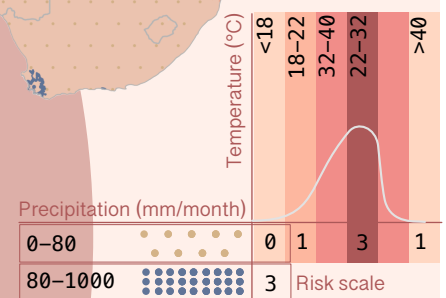
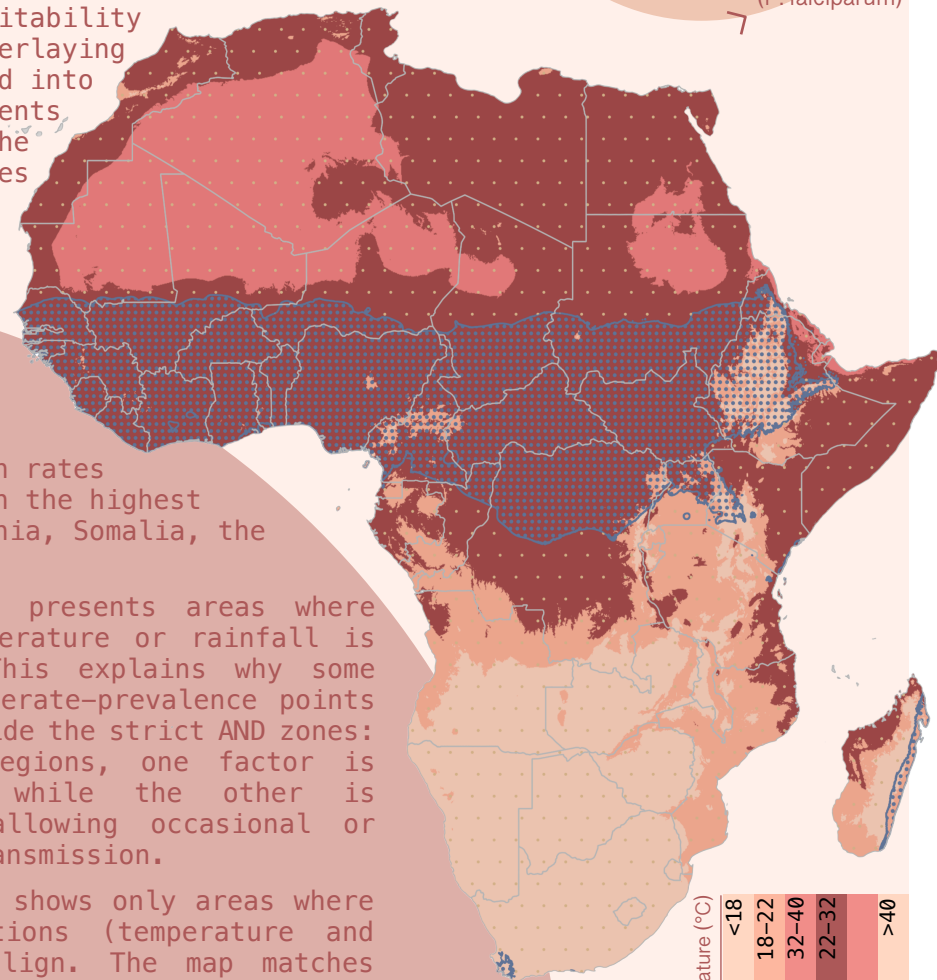
Suitability logic: AND vs OR

The distribution of *Plasmodium falciparum* prevalence shows strong spatial clustering of high infection rates across West and Central Africa, with the highest values concentrated in Kenya, Tanzania, Somalia, the DRC and Mozambique.

The OR map presents areas where either temperature or rainfall is suitable. This explains why some low- or moderate-prevalence points appear outside the strict AND zones: in these regions, one factor is favourable while the other is marginal, allowing occasional or seasonal transmission.

The AND map shows only areas where both conditions (temperature and rainfall) align. The map matches the high-PfPR point clusters, though some cases occur outside the strict suitability zones due to microclimates, seasonal mismatches, human movement, and local breeding sites not captured at this resolution.

Overall, this analysis is limited by coarse climate resolution, single-month data, and other social, ecological, and vector-specific factors not captured here.



Almost half of Africa's population is living in an area where both conditions for malaria are met.

42%

Reading the patterns, informing decisions

→ Preventive, pre-seasonal interventions in regions where suitability is consistently high

→ Research into mismatches between climatic risk and case distribution (enhanced surveillance)

→ Early warning system on climatic shifts (rainfall or temperature anomalies) to anticipate changes in vector and transmission behaviours

■ Malaria risk
■ No malaria risk

Malaria prevalence (2016) measured as *Plasmodium falciparum* infection rate

● 0 - 0,1
● 0,11 - 0,3
● 0,31 - 0,5
● 0,51 - 0,75
● 0,76 - 1